Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Chord of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ whose middle point is $\left(\frac{1}{2},\frac{2}{5}\right)$ meets the minor axis at A and major axis at B, length of AB (in units) is:

(A) $\frac{\sqrt{41}}{5}$

(B) $\frac{2\sqrt{41}}{5}$

(C) $\frac{3\sqrt{41}}{5}$

(D) $\frac{7\sqrt{41}}{5}$

Q.2

If $y=x+c$ is a normal to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, then c^2 is equal to

(A) $\frac{13}{25}$

(B) $\frac{25}{13}$

(C) $\frac{25}{9}$

(D) $\frac{13}{4}$

Q.3

If the normal at any point P on the ellipse $\frac{x^2}{64}+\frac{y^2}{36}=1$ meets the major axis at G_1 and the minor axis at G_2 then the ratio of PG_1 and PG_2 is equal to

(A) $6:8$

(B) $8:6$

(C) $9:16$

(D) $16:9$

Q.4

Where the normal is parallel to the line $4x-2y-5=0$ to the ellipse $\frac{(x^2)}{9}+\frac{(y^2)}{4}=1$ is:

(A) $\left(\frac{9}{5},\frac{8}{5}\right)$

(B) $\left(\frac{8}{5},-\frac{9}{5}\right)$

(C) $\left(-\frac{9}{5},\frac{8}{5}\right)$

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(D) $\left(\frac{8}{5}, \frac{9}{5}\right)$

Q.5

The normal at an end of a latus rectum of the ellipse $x^2/a^2 + y^2/b^2 = 1$ passes through an end of the minor axis if

(A) $e^4 + e^2 = 1$

(B) $e^3 + e^2 = 1$

(C) $e^2 + e = 1$

(D) $e^3 + e = 1$

Q.6

P is a point on the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and P' be the corresponding point on the auxiliary circle C : $x^2 + y^2 = a^2$. The normal at P to E and at P' to C intersect on circle whose radius is

(A) $a + b$

(B) $a - b$

(C) $2a$

(D) $2b$

Q.7

The normal at the point $(3, 2\sqrt{3})$ on the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ meets the major axis of the ellipse at $(\alpha, 0)$, the value of α is:

(A) $\frac{5}{3}$

(B) $\frac{3}{5}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Q.8

If CF is perpendicular from the centre of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ to the tangent at P, G is the point where the normal at P meets the major axis, then $CF \cdot PG =$

(A) 9

(B) 18

(C) 25

(D) 34

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Q.9

If $y = x$ and $3y + 2x = 0$ are the equations of a pair of conjugate diameters of an ellipse, then the eccentricity of the ellipse is

(A) $\sqrt{2/3}$

(B) $1/\sqrt{3}$

(C) $1/\sqrt{2}$

(D) $2/\sqrt{5}$

Q.10

Locus of midpoints of focal chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{ex}{a}$

(B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{ex}{a}$

(C) $x^2 + y^2 = a^2 + b^2$

(D) None of these

Q.11

If $lx + my + n = 0$ is normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then

(A) $\frac{a^2}{\ell^2} + \frac{b^2}{m^2} = \frac{\left(a^2 - b^2\right)^2}{n^2}$

(B) $\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{\left(a^2 - b^2\right)^2}{n^2}$

(C) $\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{\left(a^2 + b^2\right)^2}{n^2}$

(D) None of these

Q.12

If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$ intersects it again at the point $\mathrm{Q}(2\theta)$, then $\cos\theta$ is equal to

(A) $2/3$

(B) $-2/3$

(C) $1/3$

(D) $3/4$

Q.13

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If the chords of contact of tangents from two points (α, β) and (γ, δ) to the ellipse $\frac{x^2}{5}+\frac{y^2}{2}=1$ are perpendicular, then $\frac{\alpha \gamma}{\beta \delta}=$

(A) $\frac{4}{25}$

(B) $\frac{-4}{25}$

(C) $\frac{25}{4}$

(D) $\frac{-25}{4}$

Q.14

If the ellipse $9 x^2+16 y^2=144$ intercepts the line $3x+4y=12$, then what is the length of the chord so formed?

(A) 5 units

(B) 6 units

(C) 8 units

(D) 10 units

Q.15

If $\mathrm{P}(\theta)$ and $\mathrm{Q}\left(\frac{\pi}{2}+\theta\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then locus of the mid- point of PQ is

(A) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$

(B) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$

(C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$

(D) None of these

Answers & Solutions

Q.1 Answer:

$\frac{\sqrt{41}}{5}$

Solution:

Equation of the chord is

$\frac{1/4}{25}+\frac{4/25}{16}-1=\frac{(1/2)x}{25}+\frac{(2/5)y}{16}-1\,(T=S_1)$

$\Rightarrow 4x+5y=4$

which meets minor axis at $A\left(0, \frac{4}{5}\right)$ **and major axis at** $B(1,0)$

So the length of $AB=\sqrt{1+\frac{16}{25}}=\frac{\sqrt{41}}{5}$ **units.**

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Q.2 Answer:

$\frac{25}{13}$

Solution:

Any point on the ellipse is $P(3\cos\theta, 2\sin\theta)$ **equation of the normal at** P **is** $3.x\sec\theta - 2.y\,\text{cosec}\,\theta = 9 - 4$

Comparing it with $y = x + c$**, we get** $\frac{1}{3\sec\theta} = \frac{1}{2\,\text{cosec}\,\theta} = \frac{-c}{5}$

$\Rightarrow \quad \cos\theta = \frac{-3c}{5}, \sin\theta = \frac{-2c}{5}.$

$\Rightarrow \quad 9c^2 + 4c^2 = 25$ **as** $\cos^2\theta + \sin^2\theta = 1$

$\Rightarrow \quad c^2 = \frac{25}{13}$

Q.3 Answer:

$9 : 16$

Solution:

Equation of the normal at $P(8\cos\theta, 6\sin\theta)$ **is** $8x\sec\theta - 6y\,\text{cosec}\,\theta = 64 - 36 = 28.$

So G_1 **is** $\left(\frac{28}{8}\cos\theta, 0\right)$ **and** G_2 **is** $\left(0, \frac{-28}{6}\sin\theta\right)$

$$(PG_1)^2 = \left(8 - \frac{28}{8}\right)^2 \cos^2\theta + 36\sin^2\theta$$

$$= \frac{36}{64}\left(36\cos^2\theta + 64\sin^2\theta\right)$$

$(PG_2)^2 = 64\cos^2\theta + \left(6 + \frac{28}{6}\right)^2 \sin^2\theta$

$= \frac{64}{36}\left(36\cos^2\theta + 64\sin^2\theta\right)$

$\left(\frac{PG_1}{PG_2}\right)^2 = \frac{36}{64} \times \frac{36}{64} \Rightarrow \frac{PG_1}{PG_2} = \frac{36}{64} = \frac{9}{16}.$

Q.4 Answer:

$\left(\frac{9}{5}, \frac{8}{5}\right)$

Solution:

Slope of the normal is 2, so slope of the tangent is $-\frac{1}{2}$**.**

Equation of the tangent is $y = -\frac{1}{2}x \pm \sqrt{9 \times \frac{1}{4} + 4}$

$\Rightarrow \quad x + 2y = \pm 5$

If it touches at (x_1, y_1)**, then equation of the tangent at** (x_1, y_1) **to the ellipse is** $\frac{xx_1}{9} + \frac{yy_1}{4} = 1$

Comparing (1) and (2) we get

$$\frac{x_1}{9} = \frac{y_1}{8} = \pm\frac{1}{5}$$

$$\Rightarrow \quad (x_1, y_1) = \left(\frac{9}{5}, \frac{8}{5}\right)$$

Q.5 Answer:

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$e^4 + e^2 = 1$

Solution:

Let an end of a latus rectum be $\left(ae, b\sqrt{1-e^2}\right)$**, then the equation of the normal at this end is** $\frac{x-ae}{ae/a^2} = \frac{y-b\sqrt{1-e^2}}{b\sqrt{1-e^2}/b^2}$

It will pass through the end $(0, -b)$ **if** $-a^2 = \frac{-b^2\left(1+\sqrt{1-e^2}\right)}{\sqrt{1-e^2}}$ **or** $\frac{b^2}{a^2} = \frac{\sqrt{1-e^2}}{1+\sqrt{1-e^2}}$

or $\left(1-e^2\right)\left[1+\sqrt{1-e^2}\right] = \sqrt{1-e^2}$

or $\sqrt{1-e^2} + 1 - e^2 = 1$ **or** $e^4 + e^2 = 1.$

Q.6 Answer:

$a + b$

Solution:

$P(a\cos\theta, b\sin\theta), P'(a\cos\theta, a\sin\theta)$

Equation of the normal at P **is**

$ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$ **(1)**

Equation of the normal at P' **is**

$x\sec\theta - y\operatorname{cosec}\theta = 0$ **(2)**

From (1) and (2) we get $(a-b)y\operatorname{cosec}\theta = a^2 - b^2$

$\Rightarrow y = (a+b)\sin\theta.$

From (2) we get $x = (a+b)\cos\theta$**. So point of intersection of (1) and (2) lie on a circle of radius** $a + b.$

Q.7 Answer:

$\frac{5}{3}$

Solution:

Equation of the normal is $\frac{x-3}{3} = \frac{y-2\sqrt{3}}{\frac{2\sqrt{3}}{16}}$

$\Rightarrow 12\sqrt{3}(x-3) = 8(y-2\sqrt{3})$

which meets the major axis $y = 0$ **at point for which** $12\sqrt{3}(x-3) = -16\sqrt{3} \Rightarrow x = \frac{5}{3}$

$\Rightarrow \alpha = \frac{5}{3}$

Q.8 Answer:

9

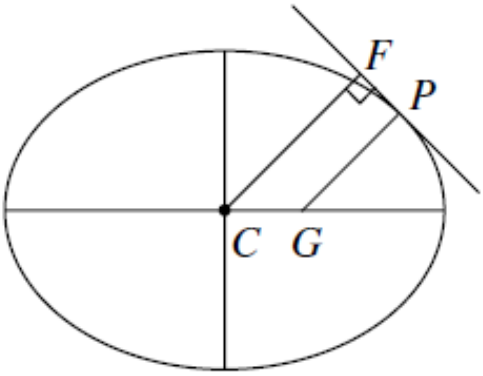
Solution:

Let $P(5\cos\theta, 3\sin\theta)$

Equation of the tangent at P **is** $\frac{x\cos\theta}{5} + \frac{y\sin\theta}{3} = 1$

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$\Rightarrow CF = \frac{1}{\sqrt{\frac{\cos^2\theta}{25} + \frac{\sin^2\theta}{9}}} = \frac{15}{\sqrt{9\cos^2\theta + 25\sin^2\theta}}$



Equation of the normal at P is

$\frac{5x}{\cos\theta} - \frac{3y}{\sin\theta} = 25 - 9 = 16$

Coordinates of G are $\left(\frac{16\cos\theta}{5}, 0\right)$

$\Rightarrow PG = \frac{3}{5}\sqrt{9\cos^2\theta + 25\sin^2\theta}$

Hence $CF \cdot PG = 9.$

Q.9 Answer:

$1/\sqrt{3}$

Solution:

Let the equation of the ellipse be $x^2/a^2 + y^2/b^2 = 1$

Slope of the given diameters are $m_1 = 1, m_2 = -2/3.$

$\Rightarrow m_1 m_2 = -2/3 = -b^2/a^2$

[using the condition of conjugacy of two diameters] $3b^2 = 2a^2 \Rightarrow 3a^2\left(1 - e^2\right) = 2a^2$

$1 - e^2 = 2/3 \Rightarrow e^2 = 1/3 \Rightarrow e = 1/\sqrt{3}$

Q.10 Answer:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{ex}{a}$

Solution:

Passing through $(ae, 0)$

$\frac{he}{a} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{xe}{a}$

Q.11 Answer:

$\frac{a^2}{\ell^2} + \frac{b^2}{m^2} = \frac{\left(a^2 - b^2\right)^2}{n^2}$

Solution:

$ex + my + n = 0$

$\alpha x \sec\theta - by\,\mathrm{cosec}\,\theta + \left(b^2 - a^2\right) = 0$

$\Rightarrow \frac{\ell}{a\sec\theta} = \frac{m}{b\cos ec\theta} = \frac{n}{b^2 - \alpha^2}$

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$\Rightarrow \cos\theta = \frac{\omega}{\ell(b^2-a^2)}, \sin\theta = -\frac{bn}{m(b^2-a^2)}$

$\Rightarrow \frac{a^2}{\ell^2} + \frac{b^2}{m^2} = \frac{\left(b^2-\alpha^2\right)^2}{n^2}$

Q.12 Answer:

$-2/3$

Solution:

Equation of normal at $(a\cos\theta, b\sin\theta)$ **is** $ax\sec\theta - by\,\mathrm{cosec}\,\theta = a^2 - b^2$

Now this normal passes through

$(a\cos 2\theta, b\sin 2\theta)$

so $aa\frac{\cos 2\theta}{\cos\theta} - bb\frac{\sin 2\theta}{\sin\theta} = a^2 - b^2$

$\Rightarrow 18\cos^2\theta - 9\cos\theta - 14 = 0 \Rightarrow \cos\theta = -\frac{2}{3}$ **or** $\cos\theta = \frac{7}{6}$ **(Not possible)**

Q.13 Answer:

$\frac{-25}{4}$

Solution:

The equation of chord of contact of tangents from two points (α, β) **and** (γ, δ) **to the given ellipse are** $\frac{x\alpha}{5} + \frac{y\beta}{2} = 1 \quad \ldots(1)$ **and** $\frac{x\gamma}{5} + \frac{y\delta}{2} = 1 \ldots(2)$

Since (1) **and** (2) **are** $\perp \therefore \frac{-2\alpha}{5\beta} \times \frac{-2\gamma}{5\delta} = -1 \Rightarrow \frac{\alpha\gamma}{\beta\delta} = -\frac{25}{4}$

Q.14 Answer:

5 units

Solution:

(a) Here,

$9x^2 + 16y^2 = 144$ **and** $3x + 4y = 12$

$\Rightarrow x = \frac{12-4y}{3}$

So, $9\left(\frac{12-4y}{3}\right)^2 + 16y^2 = 144$

On solving we get, $y = 0, 3$ **For** $y = 0; x = 4$

For $y = 3; x = 0$

$\Rightarrow$ **Length of chord** $= \sqrt{(0-3)^2 + (4-0)^2} = \sqrt{9+16}$

$= \sqrt{25} = 5$ **units**

Q.15 Answer:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$

Solution:

Clearly P **is** $(\mathrm{a}\cos\theta, \mathrm{b}\sin\theta)$ **and** $\mathrm{Q}1$ s

$(-a\sin\theta, b\sin\theta)$ **so the mid point** (h, k) **of** PQ

will be given by

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$h = \frac{a\cos\theta - a\sin\theta}{2}$ **and** $\mathrm{k} = \frac{\mathrm{b}\sin\theta + \mathrm{b}\cos\theta}{2}$

$\therefore \quad \frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 2 \Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = \frac{1}{2}$

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